

On Boundaries of Evidence

Boundary-State Calculus for Typed Observation,
Admissible Transfer, and Falsifiable Persistence

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The problem

Scientific claims often fail not within either of two descriptions, but in the passage between them: interior state to boundary data, one instrument to another, a fine model to a coarse model, a learned operator to a dynamical claim, or a topological invariant to a physical quantity. The paper develops Boundary-State Calculus (BSC) as a typed audit calculus for this passage.

BSC is not proposed as an ontology, a unified physical theory, or a proof of holography, quantum gravity, consciousness, or QCD. Its unit of evaluation is one claimed transfer. That transfer must state what moves, which observables remain meaningful, what information is destroyed, whether target equations remain satisfied, which diagrams fail to commute, what physically implements the passage, and what evidence licenses the result.

System and boundary discipline

The paper transcribes the BSC corpus's original finite-system tuple and its later measurable repair rather than replacing them retrospectively (§3). The hash-identified internal sources are not independently inspectable in the public release, so this lineage is source-local and cannot be replayed by the reader. The repaired system separates raw state from exact observational equivalence and includes controls, outputs, physical context, legal filtration, dynamics, observation, comparison geometry, viability, transformation registries, and a boundary ledger. A companion record supplies reconstruction data, provenance, and certificate state without silently changing the inherited tuple. Instrument, controller, observer or reference frame, clock, and calibration are physical configuration variables, not metadata.

An ideal trace, finite sensor layer, boundary response, reconstruction relation, and selected extension are kept distinct (§5). A boundary response supports an interior claim only relative to a governing equation, coefficient class, equivalence or gauge class, regularity, data regime, and stability theorem.

The transfer record

The central added object is (§7)

$$\mathfrak{M}_{\ell \rightarrow m} = (T_{\ell m}, T_{\ell m}^{\sharp}, K_{\ell m}, R_{\ell m}, \Theta_{\ell m}, \delta_{\ell m}, C_{\ell m}, \text{Cert}_{\ell m}).$$

$T_{\ell m}$ transports states; $T_{\ell m}^{\sharp}$ pulls target observables in the reverse direction. $K_{\ell m}$ is post-processing or a simulation channel. $R_{\ell m}$ records target-equation failure. $\Theta_{\ell m}$ is a declared naturality defect. $\delta_{\ell m}$ is directed Blackwell–Le Cam deficiency: the best error when the source experiment simulates the target. Zero deficiency means arbitrarily accurate simulation; exact Blackwell simulation additionally requires an attaining kernel or an applicable randomization theorem. $C_{\ell m}$ records carrier, controller, instrument, reference frame, clock, resources, and boundary conditions. $\text{Cert}_{\ell m}$ binds assumptions, tolerances, sources, proof or execution artifacts, hashes, open obligations, and status.

For a declared claim, admissibility requires a well-formed and evaluated record, certified upper enclosures within every applicable tolerance, true hard gates, and adequate readiness. An enclosure that straddles a tolerance is inconclusive: it blocks certification but does not prove a violation. These coordinates do not collapse into one score. Equation residual, observation defect, statistical deficiency, quotient loss, and viability failure occupy different spaces and obey different propagation laws (§8). A missing carrier or undefined observable pullback is a blocked transfer, not a small numerical error.

Composition and persistence

For compatible partial stochastic transfers, state kernels compose only on points whose intermediate law is supported in the next domain; observable pullbacks then compose in reverse through the corresponding restricted pullback. Observation defects have a contraction-weighted bound; directed deficiencies satisfy a triangle inequality; deterministic equation residuals propagate only with a compatible equation-space map (§7.1; §15). Incompatible clocks, interfaces, resources, or boundary conditions make the composite undefined.

Persistence is defined through induced path laws (§9). Endpoint return is insufficient. Every required prefix must remain within operational-identity, viability, residual, deficiency, and certificate budgets. “A persistent object is a recursively maintained finite boundary-state” is therefore a target-relative organizing principle. Its promotion to a universal physical law remains conjectural, untested, and unexecuted.

Provenance and status

- **Imported mathematics.** Markov kernels and operational quotients; Blackwell–Le Cam comparison; boundary trace and Calderón-type response; Koopman residual methods and stated learnability limits; periodic homogenization; sheaf descent; structured cospans; non-invertible defects; and proof checking. These are not claimed as new BSC discoveries.
- **Paper-level consequences.** Assumption-bound results include quotient descent; paired state–observable probability preservation; defect, deficiency, and deterministic residual propagation; prefix persistence obstruction; operational factorization and discard obstruction; a generic structured-cospan category conditional on a supplied decoration functor; dependency-DAG readiness caps; and the block from winding to physical charge. No empirical status is inherited.
- **Open work.** Universal persistence, empirical validation, stochastic residuals, instrument-aware reconstruction, QRF descent, certified recurrence, a coherent BSC-specific decoration functor, physical charge bridges, and a machine-checked kernel remain conjectural, untested, unexecuted, or open.

Eight fixtures

The exact fixtures (§16) are:

1. torus winding $(2, -3)$, with electric charge blocked without a physical bridge;
2. distinct one-dimensional conductivities with one Dirichlet-to-Neumann map;
3. binary experiments with directed deficiencies 0 and $1/4$;
4. Koopman residual $\sqrt{2/N}$ separated from polluted zero eigenvalues;
5. two invariant, correlated Bell states with common reduced state $I/2$ whose finite $\mathbb{Z}_2$ frame transforms reduce to orthogonal outputs;
6. locally nonempty parity constraints with no global section;
7. the off-shell residual m^2c of a massive-field shift; and
8. an exact receipt for $x = -1$ refuting $\sqrt{x^2} = x$.

Fixtures 1–7 have mathematical derivations but no separate execution receipts. Fixture 8 has a deterministic CPython receipt. The package is a preprint with audit artifacts, not a complete reference implementation.

Physical promotion and the test

Winding and physical charge are different typed objects (§13). A charge claim needs a derived bridge from a gauge representation, action and current, Gauss law, anomaly condition, boundary condition, or calibrated measurement. Primordial-field toroidal models are therefore treated as candidate topological and stabilization grammar, not an established derivation of quark charge, QCD, or hadron dynamics (§17).

Each claim has a local mathematical verdict—ill-posed, open, true, or false—and five readiness coordinates: mathematical support, empirical, computational, source, and transfer (§2.2). Dependency edges cap named readiness coordinates; they never propagate verdicts. A theorem does not raise empirical status; a hash establishes identity rather than truth; a proof DAG is not a proof; and an unexecuted calculation remains unexecuted.

The appropriate test is adversarial and narrow: Does an equivalent compositional record already exist? Is a map ill-typed? Does a composition law fail? Is there a counterexample to a fixture? Does the record change no scientific decision? If existing formalisms already supply the same obligations and readiness-cap semantics, or if BSC never separates an invalid transfer from a nearby valid one, the added structure is unnecessary. If it exposes that distinction and preserves it through composition, that—not a claim about the universe—is the contribution requiring further work.